

Mathematical Explanation for NYC Ride Demand Forecasting in 2025

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Abstract

The following is the mathematical formulas in the model for the forecasting of hourly Uber and Lyft ride demand in New York City. Before building the forecasting model, large-scale trip records are cleaned, transformed, and aggregated using DuckDB with SQL and PowerBI, and are integrated with weather and geographic information. The forecasting framework incorporates temporal features, including lagged demand, as well as weather and location-based predictors to capture spatial and temporal variation in ride demand. The project aims to evaluate the predictive performance of different modeling approaches and identify key factors associated with changes in ride demand across New York City.

1 Overview of Final Mathematical Structure for Modeling

z = pickup zone, c = company, t = hour index, k = lag index, $\epsilon_{z,c,t}$ = unobserved variation

$$\hat{D}_{z,c,t} = f \left(\underbrace{T_t}_{\text{Time}}, \underbrace{W_t}_{\text{Weather}}, \underbrace{D_{z,c,t-k}}_{\text{Lag}}, \underbrace{\bar{D}_{z,c,t}}_{\text{Rolling}}, \underbrace{S_z}_{\text{Spatial}}, \underbrace{C_c}_{\text{Company}} \right) + \epsilon_{z,c,t}$$

2 Target Variable and Lag features

$Y_{z,c,t}$ = Demand of zone z , company c , at hour t

$$Lag1_t = Y_{t-1}$$

$$Lag2_t = Y_{t-2}$$

$$Lag24_t = Y_{t-24}$$

$$Lag168_t = Y_{t-168} (168 = 7 * 24 = 1 \text{ week})$$

3 Rolling features

$$MA_{3,t} = \frac{Y_{t-1} + Y_{t-2} + Y_{t-3}}{3}$$

$$MA_{24,t} = \frac{1}{24} \sum_{i=1}^{24} Y_{t-i}$$

4 Predictor Vector

$$X_{z,c,t} = [T_t, W_t, L_{z,c,t}, R_{z,c,t}, S_z, C_c]$$

$$L_{z,c,t} = [Y_{t-1}, Y_{t-2}, Y_{t-24}, Y_{t-168}]$$

$$R_{z,c,t} = [MA_{3,t}, MA_{24,t}]$$